



The toxicity and efficacy evaluation of different fractions of Kansui fry-baked with vinegar on Walker-256 tumor-bearing malignant ascites effusion rats and normal rats



Jian-Wei Lou^a, Liang-Liang Cao^{a,b}, Qiao Zhang^a, Dong-jing Jiang^a, Wei-Feng Yao^a, Bei-Hua Bao^a, Yu-Dan Cao^a, Yu-Ping Tang^c, Li Zhang^{a,*}, Kun Wang^a, Guan-Cheng Dai^a

^a Jiangsu Key Laboratory for High Technology Research of TCM Formulae, National and Local Collaborative Engineering Center of Chinese Medicinal Resources Industrialization and Formulae Innovative Medicine and Jiangsu Collaborative Innovation Center of Chinese Medicinal Resources Industrialization, Nanjing University of Chinese Medicine, Nanjing 210023, China

^b Department of Pharmacy, Nanjing Hospital of Traditional Chinese Medicine, Nanjing 210001, China

^c College of Pharmacy and Shaanxi Collaborative Innovation Center of Chinese Medicinal Resources Industrialization, Shaanxi University of Chinese Medicine, Xi'an 712046, Shaanxi Province, China

ABSTRACT

Ethnopharmacological relevance: Kansui, the root of *Euphorbia kansui* S.L.Liou ex S.B.Ho (*E.kansui*), is a classical traditional Chinese medicine (TCM) with certain toxicity. According to the theory of TCM, kansui fry-baked with vinegar (VEK) possesses low toxicity and mild diuretic and purgative efficacy. In clinical practice, it is commonly used for the treatment of ascites and oliguria. The present study aimed to evaluate the toxicity and efficacy of different fractions of VEK and reveal the underlying material basis by employing an animal model of malignant ascites effusion (MAE) in rats.

Materials and methods: The MAE rats as the model were constructed in SPF male wistar rats by intraperitoneal injection of Walker-256 tumor cells. The MAE rats were used and randomly divided into the control group (normal rats), control groups with different fractions (VEKA, VEKB, VEKC and VEKD), model group (MAE rats), positive control group (model group with furosemide), model groups with different fractions (VEKA, VEKB, VEKC and VEKD). Histopathological observation was used to confirm Walker-256 tumor-bearing organ injuries in rats. For the efficacy evaluation, the ascites and urine volumes, the urinary electrolyte concentrations (Na^+ , K^+ and Cl^-) and pH, the ascites levels of pro-inflammatory cytokines (IL-2, IL-6, TNF- α , IFN- γ and VEGF), PRA, the serum levels of Ang II, ALD and ADH, as well as AQP8 protein expression in the gastrointestinal tract were detected. Furthermore, different levels of indicators were measured in the toxicity evaluation of different fractions both on normal and model rats, including serum liver enzymes (AST and ALT), serum oxidative damage parameters (GSH, MDA, LDH and SOD), expressions of inflammatory parameters (NF- κ B, ICAM-1 and E-cadherin) and apoptosis signals (caspase-3, -8, -9, Bcl-2 and Bax) in the liver and gastrointestinal tract.

Results: Walker-256 tumor-bearing malignant ascites effusion rats showed obvious hepatic and gastrointestinal injuries by histopathological observation. In the efficacy evaluation, model rats treated with VEKB and VEKC showed significant urine increase (VEKB, $P < 0.01$; VEKC, $P < 0.01$) and ascites reduction (VEKB, $P < 0.01$; VEKC, $P < 0.01$). These two fractions also balanced the concentrations of Na^+ , K^+ and Cl^- in urine (VEKB, all $P < 0.05$; VEKC, all $P < 0.05$), remarkably decreased urinary pH (VEKB, $P < 0.01$; VEKC, $P < 0.01$), and reduced the ascites levels of IL-2, IL-6, TNF- α , IFN- γ and VEGF (VEKB, all $P < 0.01$; VEKC, all $P < 0.01$) in the model rats. Moreover, levels of PRA, the serum Ang II, ALD and ADH of model rats were decreased after treated by VEKB and VEKC (VEKB, all $P < 0.05$; VEKC, all $P < 0.05$). Meanwhile, the expression of gastrointestinal AQP8 of the model rats was also enhanced after treated by VEKB and VEKC (VEKB, $P < 0.01$; VEKC, $P < 0.01$). In the toxicity evaluation, although VEKB and VEKC caused toxic indexes moved to the worse aspects in normal rats, nearly all of these indicators notably improved in the model rats. Additionally, VEKA showed no effect on the indicators, either in the efficacy evaluation or in the toxicity evaluation. And VEKD could significantly improve some indicators (urine volume, concentration of K^+ in urine, serum MDA, AI and caspase-9) in MAE rats.

Conclusions: VEKB and VEKC were demonstrated a significant efficacy in treating malignant ascites effusion,

* Corresponding author.

E-mail address: zhangli@njucm.edu.cn (L. Zhang).

which could reduce hepatic and gastrointestinal damage on the model rats but cause the same damage to the normal. These data embody the traditional Chinese medicine application principle: *You Gu Wu Yun*. And these results will provide reference for the safer and better clinical utilization of kansui.

1. Introduction

You Gu Wu Yun (Chinese: 有故无殒), a clinical application principle for the toxic traditional Chinese medicine (TCM), covers the systematic and complex interconnecting among the disease, medicine and treatment (Peng et al., 2015). Briefly, the toxicity of the poisonous TCM should be dialectical treated. For instance, hypertoxic Arsenic shows miraculous anticancer activity in patients with acute promyelocytic leukemia (Zhang et al., 2010; Zhu and Huang, 2014).

Kansui, the dried root of *Euphorbia kansui* S.L.Liou ex S.B.Ho (*E.kansui*), is an effective and commonly used traditional Chinese medicine, which is widely applied for centuries in China as a remedy for edema, ascites and pleural fluid effusion, and difficulty in urination and defecation (Commission, 2015). Recently, its potential curative effects on leukemia (Wu et al., 1991), HIV (Cary et al., 2016; Fujiwara et al., 1998), cancer (Ma et al., 2016; Miyata et al., 2004; Yasukawa et al., 2000; Yoshida et al., 2010), pancreatitis (Lu et al., 2007, 2004; Yi et al., 2007) and regulation of the immune system (Nunomura et al., 2006) having been reported. However, a series of studies have found the toxic side effects of kansui, such as organ toxicity (Gao et al., 2015; Yan et al., 2014, 2016; Zhang et al., 2012), skin irritation (Diao, 2007), proinflammation (Shu et al., 2010) and tumor promotion (Matsumoto et al., 1992), which having seriously restricted its clinical applications.

Processing of toxic TCM, very special technique of “Pao Zhi”, should be considered seriously since it has the important influence on the TCM clinical safe applications. For example, kansui have to be fry-baked with vinegar for oral administration, which was stipulated in the Chinese Pharmacopoeia (Commission, 2015). This herbal TCM has been utilized as an effective medicine to ascites effusion for centuries in China. Cao et al. (2015b) found although kansui fry-baked with vinegar (VEK) showed some degree of organic toxicity on the normal rats, also reduced the hepatic and gastrointestinal injuries in malignant ascites effusion (MAE) rats. Additionally, Zhang et al. (2016) found that VEK could obviously reduce ascites and increase urine on the MAE rats. However, the material basis of VEK's toxicity and efficacy on MAE is still unclear.

In this study, the toxicity and efficacy of different fractions of kansui fry-baked with vinegar on normal and MAE rats were evaluated, and the application of *YOU GU WU YUN* was tentatively discussed.

2. Materials and methods

2.1. Plant materials and preparation of drug's fractions

The roots of *E.kansui* were collected in April 2015 from Baoji, Shaanxi Province, China. The crude drug was identified by Professor Yuping Tang and the voucher specimen (No. NJUTCM-20100520 for *Euphorbia kansui*) has been deposited in the Herbarium of Nanjing University of Chinese Medicine.

The preparation of different fractions of kansui fry-baked with vinegar (VEKF) were conducted according to the following procedures: Firstly, the dried and cleaned raw kansui (10.0 kg) mixed evenly with vinegar (3.0 kg) in a stainless steel barrel. The barrel was sealed until the vinegar was fully absorbed. After that, the herbal material was fry-baked (260 °C, 9 min) until the color deepened and scorched spots appeared slightly. After cooling down, the air-dried and powdered drug was exhaustively extracted by successive refluxing with 95% EtOH (keep faint boiling, 2 h) two times. The residue was exhaustively extracted by the same way with water (keep faint boiling, 2 h) two times. Then we acquired the compound extract (1.1 kg) by mixing the two

extracts as above, and the solvent was removed under reduce pressure. After that, the concentrated total extract was partitioned with ethyl acetate and the concentrated EtOAc extract (226.4 g, 2.26%) was subjected to silica gel column chromatography (10 cm × 300 cm i.d.; 2000 g, 200–300 mesh) under hypobaric condition. Elution with 20.0 L petroleum ether-ethyl acetate (100:2) yielded fraction VEKA (55.7 g, 0.55%), next elution with 8.0 L petroleum ether-ethyl acetate (100:15) and 2.0 L petroleum ether-ethyl acetate (100:18) yielded fraction VEKB (21.3 g, 0.21%). Then elution with 8.0 L petroleum ether-ethyl acetate (100:40) and 2.0 L petroleum ether-ethyl acetate (100:50) yielded fraction VEKC (32.5 g, 0.32%), finally, elution with methyl alcohol yielded fraction VEKD (155.6 g, 1.5%). All the foregoing approximate yields were based on the weight of the crude drug. And the dry fractions were resuspended in 0.5% CMC-Na for oral administration.

Additionally, an UFLC-Q-TOF/MS method was developed to determine the chemical profile of each fraction (Lou et al., 2017). And the results showed that 22 characteristic diterpenoids were identified tentatively by comparing their accurate mass, corresponding fragmentation ions with studies of *E.kansui*. The detailed protocol and structure information were depicted in [supplementary materials](#).

2.2. Chemicals and reagents

HPLC/MS-grade acetonitrile and formic acid (Merk, Darmstadt, Germany) were used for UFLC-Q-TOF/MS analysis. The pure water for the UPLC analysis was purified using a Milli-Q water purification system (Millipore, MA, USA). Regent-grade ethanol (Hanbang, Huaian, China), petroleum ether (Xilong, Shantou, China) and ethyl acetate (Xilong, Shantou, China) were used for extraction and separation.

Sodium carboxymethyl cellulose (CMC-Na), chloral hydrate and NaCl were purchased from Sinopharm Chemical Reagent Co., Ltd (Shanghai, China). The Interleukin-2 and 6 (IL-2 and 6), Tumor necrosis factor alpha (TNF-α), Interferon gamma (IFN-γ) and Vascular endothelial growth factor (VEGF) enzyme-linked immunosorbent assay (ELISA) kits were purchased from eBiosciences, Inc., USA. The blood biochemistry [Alanine aminotransferase (ALT), Aspartate aminotransferase (AST), Glutathione (GSH), Malondialdehyde (MDA), Lactate dehydrogenase (LDH), Superoxide dismutase (SOD), Plasma Renin Activity (PRA), Angiotensin II (Ang II), Aldosterone (ALD) and Anti diuretic hormone (ADH)] detection kits, Na⁺, K⁺, Cl⁻ detection kits, apoptosis in situ detection kit, primary antibodies to polyclonal anti-NF-κB p65, anti-ICAM-1, anti-E-cadherin, anti-caspase-3, anti-caspase-8, anti-caspase-9, anti-Bcl-2, anti-Bax, anti-Aquaporin 8, hematoxylin and eosin were all purchased from Nanjing JianCheng Institute of Biotechnology (Nanjing, China). And the DAB kit was purchased from Fuzhou new biotechnology Co. Ltd (Shanghai, China). TRIzol was purchased from Thermo Fisher Scientific, Inc (Waltham, Massachusetts, USA). Real-Time PCR Master Mix (SYBR Green) was purchased from TOYOBO (Shanghai) biotechnology co., Ltd. PrimeScript RT reagent kit was purchased from Treasure biological engineering (Dalian) Co., Ltd. The primers used in RT-PCR were purchased from RuiJinSi technology Co., Ltd (Nanjing, China). PBS and normal goat serum were purchased from Beijing Solarbio Science and Technology Co., Ltd (Beijing, China).

2.3. Animals and treatment

SPF male wistar rats were purchased from Shipper B&K Laboratory Animal Technology Corp., Shanghai, China (license number: SCXK (Zhejiang) 2014-0001). All of the rats were kept in Drug Safety Evaluation Center of Nanjing University of Chinese Medicine (Nanjing,

China). All of the studies on animals were in accordance with the guidelines of the Animal Ethics Committee of Nanjing University of Chinese Medicine.

After a week of adaptive feeding, eighty-eight rats weighing 210–250 g were randomly divided into eleven groups (8 rats in each group): control group (normal rats), model group (the malignant ascites effusion rats), positive control group (model group with furosemide), control and model groups with different VEKF (VEKA, VEKB, VEKC and VEKD). The malignant ascites effusion (MAE) rats as the model were constructed in SPF male Wister rats by reference to previous method (Shen et al., 2016), and the difference is the inoculation of tumor cells by intraperitoneal injection. On the next day with establishment of model, the rats of the different groups were given respective drugs by gavage for 7 days. The control group and model group were given 0.5% CMC-Na, and the positive control group was given 4 mg/kg of furosemide. The VEKF supplemented groups were given the corresponding VEKA, VEKB, VEKC and VEKD suspensions respectively (all 0.1 mL/10 g). And the doses of all the drugs were listed in Table 1. The different dosage is achieved by multiplying the effective dose of kansui fry-baked with vinegar (VEK) by the different foregoing approximate yields (VEKA, 0.55%; VEKB, 0.21%; VEKC, 0.32%; VEKD, 1.5%) of VEKF. VEK could effectively promote urination and reduce the malignant ascites in malignant ascites effusion (MAE) rats at a dose of 340 mg/kg (Cao et al., 2015a). Based on the preliminary experiment, the dose of 680 mg/kg (VEK, about eight times clinical dose) was finally used to calculate the different dosage.

For the purpose of twenty-four hours urine collection, the rats were weighed and housed in metabolism cages on the 6th day after administration. The urine samples were stored at -20°C until the examination of pH, Na^{+} , K^{+} and Cl^{-} levels. When the administration of drug was over, each rat was weighed and anesthetized with 10% chloral hydrate by intraperitoneal injection before the ascites was carefully collected. All the ascites samples were stored at -20°C until the examination of pro-inflammatory cytokines. The blood samples were collected from the abdominal aorta of rats which were fasted for 12 h, subsequently subjected to centrifugation (3500 rpm at 4°C for 10 min) and were stored at -80°C for detection of biochemical parameters. The liver, stomach and small intestine were also removed from rats, being washed in a saline solution (0.9% NaCl). A part of the tissues were fixed in 4% neutral paraformaldehyde, the others were stored at -80°C until other assays.

2.4. Histopathological observation of Walker-256 tumor-bearing organ injuries in rats

The liver, stomach and small intestine tissues were stained with hematoxylin-eosin by the Department of Pathology, Jiangsu Province Hospital of TCM. The slices were imaged with a microscope (magnification, 100 $\times$) for pathological differences observation.

2.5. Efficacy evaluation of VEKA, VEKB, VEKC and VEKD

For purpose of efficacy evaluation, some indexes of rat urine and ascites were analyzed. Additionally, the levels of PRA, serum Ang II, ALD and ADH, the expression of AQP8 protein and its mRNA in the stomach and small intestine were also be analyzed.

2.5.1. Determination of the volumes of ascites and urine

As mentioned above, the volume of twenty-four hours urine samples was regarded as the urine volume. In order to minimize the error during the sample collection, all the metabolism cages were carefully washed prior to formal collection. The ascites was collected before all the weighed rats were anesthetized. The animals were fixed in the supine position, the peritoneal cavity was opened carefully, and the ascites was removed using an injector and drinking paper. Additionally, the rats were weighed again, and the difference in the weight was regarded as

the ascites volume.

2.5.2. Assessment of urinary electrolyte concentrations and pH

Urine samples of each pair of rats were taken out from the environment at -20°C . After that, the urinary Na^{+} , K^{+} , and Cl^{-} levels were measured according to the manufacturer's specification. Urinary pH was directly determined on fresh urine samples using HIP 3221 (Hanna Instruments) pH meter.

2.5.3. Level determination of IL-2 and 6, TNF- α , IFN- γ and VEGF in ascites

When the ascites was achieved, the pro-inflammatory cytokines which including IL-2 and 6, TNF- α , IFN- γ and VEGF were quantified by utilizing ELISA according to the manufacturer's specification. The results were expressed as nanogram per milliliter ascites.

2.5.4. Level determination of PRA, serum Ang II, ALD and ADH

When the serum was prepared, ADH and the main mediators in the rein-angiotensin-aldosterone system (RAAS) which including PRA, Ang II and ALD were quantified by using commercially detection kits. And all the steps during the experiment were following standard operating procedures. The results were expressed as nanogram per milliliter serum.

2.5.5. IHC of AQP8 and Real-time PCR analysis of the expression of AQP8 mRNA

The expression of AQP8 protein in the stomach and small intestine tissues was detected by immunohistochemical analysis. The expression of AQP8 mRNA was analyzed by Real-time PCR. Mainly including: First: total RNA from frozen rats' tissues (the stomach and small intestine) was extracted using TRIzol RNA Extraction Reagent (Invitrogen, America) according to the standard protocol. Second: The first-stand cDNA was reverse transcribed from total RNA (2 μg) by PrimeScript RT reagent kit (Takara, Japan). The target mRNA expression was quantified with Real-Time PCR Master Mix (SYBR Green) (TOYOBO, Japan) using StepOnePlus™ Real-Time PCR system (Warrington, UK). The primers used were as follows: AQP8 5'-CTGGC TACTGGGACTTCCAT-3' (forward) and 5'-CTTTAGAATCAGGCGGG TTT-3' (reverse), amplicon 116 bp; GAPDH 5'-TCAAGAAGGTGGTGAA GCAG-3' (forward) and 5'-AGGTGGAAGAATGGGAGTTG-3' (reverse), amplicon 111 bp. The conditions of analysis were as follows: (1) initial activation: 95°C , 5 min; (2) cycling stage: 95°C , 15 s; 60°C , 20 s; 72°C , 40 s; 40 cycles; (3) melt curve stage: 95°C , 15 s; 60°C , 15 s; 95°C , 15 s. GAPDH was used as an endogenous control for normalization and quantification of target gene expression. The relative quantification of target gene expression was calculated using the $2^{-\Delta\Delta\text{Ct}}$ method

Table 1

The dose of different groups.

Groups	n	Dose(mg/kg)	Drugs
Control	8	–	–
Model	8	–	–
Positive control	8	–	–
VEKAC	8	3.740	VEKA
VEKBC	8	1.428	VEKB
VEKCC	8	2.176	VEKC
VEKDC	8	10.540	VEKD
VEKAM	8	3.740	VEKA
VEKBM	8	1.428	VEKB
VEKCM	8	2.176	VEKC
VEKDM	8	10.540	VEKD

VEKA, VEKB, VEKC and VEKD: fractions A, B, C and D of kansui fry-baked with vinegar.

VEKAC, VEKBC, VEKCC and VEKCD: Normal rats supplemented with VEKA, VEKB, VEKC and VEKD.

VEKAM, VEKBM, VEKCM and VEKDM: Model rats supplemented with VEKA, VEKB, VEKC and VEKD.

(Schmittgen and Livak, 2008).

2.6. Toxicity evaluation of VEKA, VEKB, VEKC and VEKD on normal and MAE rats

For the toxicity evaluation, the serum samples were prepared before the detection of liver enzymes (ALT and AST), oxidative damage indexes (GSH, MDA, LDH and SOD), while the paraffin Section (4- μ m thickness) of the liver, stomach and small intestine were prepared according to a standard protocol (Fischer et al., 2008). The inflammatory parameters (NF- κ B, ICAM-1 and E-cadherin), apoptosis signals (caspase-3, caspase-8, caspase-9, Bcl-2 and Bax) were analyzed by immunohistochemistry (IHC) on serial paraffin sections while the apoptosis index (AI) were evaluated by the TUNEL method.

2.6.1. Level determination of AST, ALT, GSH, MDA, LDH and SOD in serum

The activity of serum biochemical indexes which including ALT, AST, GSH, MDA, LDH and SOD were detected by using an automatic biochemical analyzer (AU 680; Beckman Coulter, Inc. USA). All operations in the experimental process were according to the manufacturer's specification.

2.6.2. TUNEL assay and Immunohistochemistry

2.6.2.1. TUNEL assay. TUNEL staining was performed with the Apoptosis in situ Detection Kit according to the manual protocols. The TUNEL reaction was visualized by using DAB and counterstained with hematoxylin. Nuclei were stained with DAB for the assessment of positive TUNEL reaction. Five randomly selected fields were assessed in each section. The apoptosis index was expressed as the percentage of apoptotic positive nuclei among at least 100 cells counted under a light microscope (magnification, 400 $\times$).

2.6.2.2. Immunohistochemistry. The immunohistochemistry (IHC) was conducted on prepared paraffin sections deparaffinized in xylene and re-hydrated in alcohol gradients. And the slides were washed for 3

times with the phosphate buffer saline (PBS; pH7.4). In order to retrieve the antigens, the sides were heated on water bath (keep faint boiling, 20 min) with respective retrieval solutions. The activity of endogenous peroxidase was treated with 3% H₂O₂-MeOH and blocked (room temperature, 10 min). After preincubation with normal goat serum (DAKO, Denmark) at room temperature, the slides were incubated with the appropriate primary antibodies at 37 °C for 2 h. Subsequently, the sections were washed with PBS for 3 times, incubated with the rabbit anti-mouse IgG at 37 °C for 30 min. The IHC reactions were developed with diaminobenzidine tetrahydrochloride (DAB). Finally, the slides were rinsed in distilled water, counterstained with hematoxylin, dehydrated in alcohol gradients, passed through xylene and sealed. In negative controls, PBS was used instead of primary antibody. Ten randomly selected fields were assessed in each section. For quantitative analysis of all data, the integral optical density (IOD) of target signals in the cytoplasm and (or) nucleus was performed and analyzed by Image-Pro Plus 6.0 (Media Cybernetics, USA).

2.7. Statistical analysis

SPSS 19.0 was used for statistical analysis, and all of the experimental data were expressed as mean $\pm$ standard error (SD). One-way analysis of variance (ANOVA) followed by a Fisher's Least Significant Difference (LSD) test using the GraphPad Prism version 5 for Windows (trial version, GraphPad Software, San Diego, CA, USA), was used for comparison between the groups. It was considered to be statistically significant when *p* value less than 0.05 or 0.01.

3. Results

3.1. Organ injuries of MAE rats on histopathological changes

The organ injuries of MAE rats were confirmed by histopathological examinations (Fig. 1). The MAE rats displayed massive mononuclear cell infiltration or necrocytosis of the liver, stomach and small intestine tissues. Thus, there were obvious hepatic and gastrointestinal injuries in

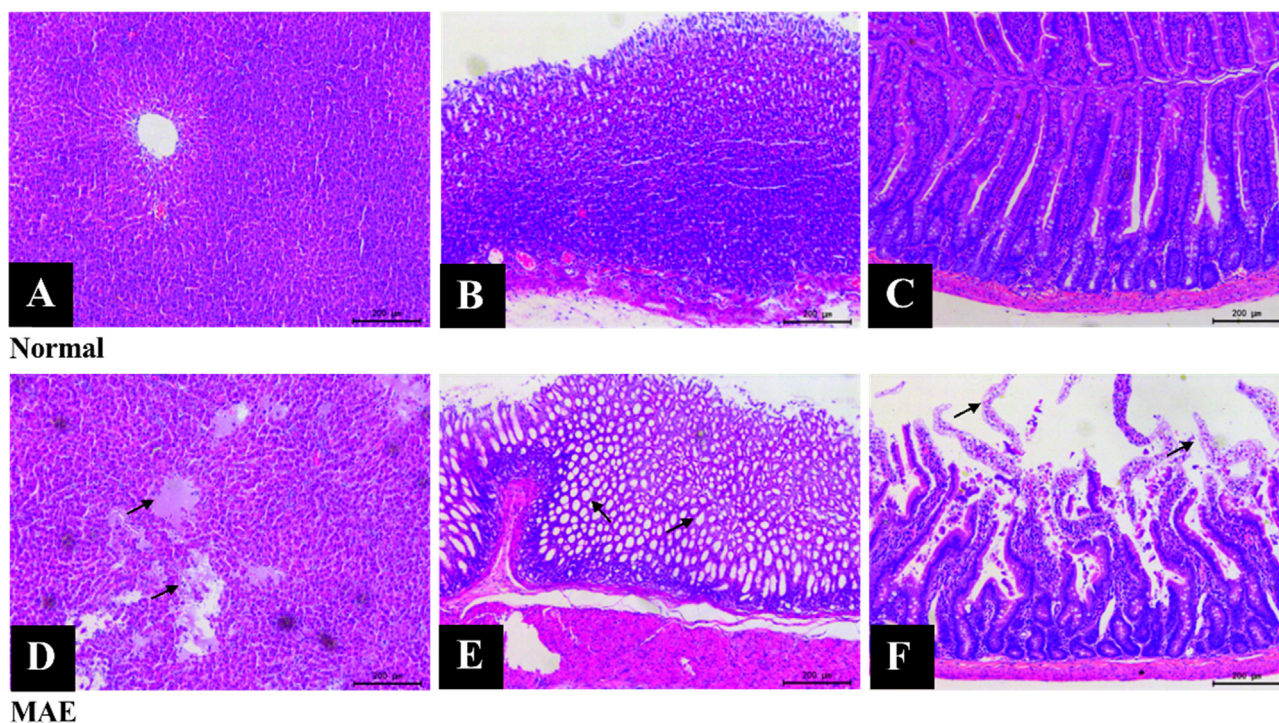


Fig. 1. Histopathological observation of Walker-256 tumor-bearing organ injuries in rats. (A, D) Liver, (B, E) Stomach, (C, F) Small intestine. Photographs of the sections (magnification, 100 $\times$) are illustrated. The cellular infiltration or necrocytosis was indicated with arrows.

model rats.

3.2. Efficacy evaluation of VEKF

3.2.1. Effects of VEKF on rat ascites and urine volume

As shown in Table 2, the model rats showed oliguria with substantive ascites. Compared with the model group, the urine volume of MAE rats was significantly increased following the administration of VEKB ($P < 0.01$) and VEKC ($P < 0.01$), while the ascites volume was decreased remarkably ($P < 0.01$). And the urine of VEKDM group also increased remarkably ($P < 0.05$) compared with that of the model group. Thus, VEKB and VEKC might play critical roles in the treatment of ascites and difficulty in urination.

3.2.2. Effects of VEKF on rat urinary electrolyte concentrations and pH

As shown in Table 2, the urinary electrolyte concentrations (Na^+ , K^+ and Cl^-) and pH in the model group were significantly higher than that in the control group ($P < 0.01$). And the levels of these factors of MAE rats were significantly decreased with the treatment of VEKB (pH, Na^+ , and Cl^- $P < 0.01$; K^+ , $P < 0.05$) and VEKC ($P < 0.05$). Under the treatment of VEKD, the concentration of K^+ in urine was remarkably improved ($P < 0.05$) in the model group. These results indicated that VEKB and VEKC might play major roles in the treatment of urinary electrolyte and acid-basic disturbances in MAE rats.

3.2.3. Effects of VEKF on pro-inflammatory cytokines in rat ascites

The ascites levels of pro-inflammatory cytokines IL-2, IL-6, TNF- α , IFN- γ and VEGF were measured to assess the severity of inflammation by respective ELISA kits (Fig. 2). Compared with the model group, these ascites pro-inflammatory cytokines in model rats were significantly down-regulated with the treatment of VEKB ($P < 0.01$) and VEKC ($P < 0.01$). The results showed that VEKB and VEKC could palliate symptoms of inflammatory in model rats.

3.2.4. Effects of VEKF on rat PRA, the serum Ang II, ALD and ADH

As shown in Table 3, PRA, the serum Ang II, ALD and ADH levels in the model group were remarkably increased compared to that of control group ($P < 0.01$). Compared with the model group, these serum indexes levels in model rats were significantly decreased under the treatment with VEKB (PRA and ADH, $P < 0.05$; Ang II and ALD, $P < 0.01$) and VEKC ($P < 0.01$). The results indicated that VEKB and VEKC could suppress the RAAS (rennin-angiotensin-aldosterone system) and the release of ADH.

3.2.5. Effects of VEKF on the expression of AQP8 and its mRNA in rat tissues

Expression of AQP8 protein in the stomach and small intestine in all seven groups was established by quantitative immunohistochemistry, and Real-time PCR of mRNA (Fig. 3). The detailed information of IHC is shown in Table S1 and Fig. S1. In these two tissues, the expression of AQP8 and its mRNA in model rats were remarkably lower compared to that of control group ($P < 0.01$). Compared with the model group, the expression of AQP8 and its mRNA in two tissues of MAE rats were significantly enhanced under the treatment with VEKB ($P < 0.01$) and VEKC ($P < 0.01$). The results demonstrated that VEKB and VEKC could regulate the abnormal expression of AQP8 protein in the gastrointestinal tract of MAE rats.

3.3. Toxicity evaluation of VEKF on normal and MAE rats

3.3.1. Effects of VEKF on rat serum AST, ALT, GSH, MDA, LDH and SOD

3.3.1.1. Effects of VEKF on the liver enzymes in rats. Table 4 shows the serum liver enzymes levels of experimental rats from different groups. The serum ALT and AST levels of rats in the VEKBC group were increased significantly compared to that of control group ($P < 0.01$). And in the group VEKCC, the level of serum ALT ($P < 0.01$) was

improved considerably. The data indicated that VEKB and VEKC might have liver toxicity. Additionally, the serum ALT and AST levels were increased dramatically in the model group than that in the control group ($P < 0.01$). The hepatic injury in model rats was further confirmed. These serum liver enzymes were improved after drugs treatment. Compared with the model group, the levels of two serum liver enzymes in MAE rats were markedly decreased under the treatment with VEKB ($P < 0.01$) and VEKC ($P < 0.01$). The results demonstrated that hepatotoxic VEKB and VEKC may not exacerbate but alleviate the liver injury in model rats.

3.3.1.2. Effects of VEKF on the oxidative damage in rats. Table 5 shows the serum parameters levels of oxidative damage in experimental rats from different groups. Compared with the control group, the serum MDA and LDH levels of rats in the VEKBC group were increased considerably ($P < 0.01$), while the levels of serum GSH ($P < 0.05$) and SOD ($P < 0.01$) were significantly lowered. And in the group VEKCC, the serum MDA level was increased markedly ($P < 0.05$), and the serum SOD level was reduced significantly ($P < 0.01$). The data indicated that VEKB and VEKC could cause oxidative damage in rats. Compared with the control group, the serum MDA and LDH levels of model rats were increased significantly ($P < 0.01$), while the levels of serum GSH and SOD were significantly decreased ($P < 0.01$). The data indicated that the successful establishment of MAE rats would with the oxidative damage. After treatment, these biochemical indexes were improved. Compared with the model group, the serum GSH and SOD levels in MAE rats were increased significantly under the treatment with VEKB (GSH, $P < 0.05$; SOD, $P < 0.01$) and VEKC ($P < 0.01$), while the serum MDA and LDH levels were remarkably decreased ($P < 0.01$). Compare with the model group, the serum MDA level in the VEKDM group was decreased significantly ($P < 0.05$). The results showed that VEKB and VEKC could cause oxidative damage, but also alleviate this injury in model rats.

3.3.2. Effects of VEKF on the inflammatory damage in rat tissues

The levels of inflammatory parameters NF- κ B, ICAM-1 and E-cadherin were measured by immunohistochemistry (Fig. 4) to analyze the effect of VEKF on the inflammatory damage in the liver, stomach and small intestine. The detailed information is shown in Table S3–S5 and

Table 2

Effects of VEKF on the volume of ascites and urine, urinary electrolyte concentrations (Na^+ , K^+ , Cl^-) and pH (mean $\pm$ SD, $n = 8$).

Groups	Ascites (g)	Urine (mL)	Na^+ (mmol/L)
Control	–	26.77 $\pm$ 7.42	22.48 $\pm$ 6.15
Model	25.70 $\pm$ 2.06	9.36 $\pm$ 2.51**	71.64 $\pm$ 3.19**
Positive control	14.70 $\pm$ 3.04##	21.19 $\pm$ 3.56##	33.94 $\pm$ 9.52##
VEKAM	22.10 $\pm$ 2.78	11.62 $\pm$ 2.55	62.47 $\pm$ 11.64
VEKBM	15.52 $\pm$ 2.81##	19.18 $\pm$ 5.78##	51.97 $\pm$ 4.69##
VEKCM	15.47 $\pm$ 2.77##	22.72 $\pm$ 3.91##	55.87 $\pm$ 7.81##
VEKDM	23.35 $\pm$ 3.31	14.50 $\pm$ 5.62#	72.14 $\pm$ 8.07

Groups	K^+ (mmol/L)	Cl^- (mmol/L)	pH
Control	159.21 $\pm$ 13.28	133.18 $\pm$ 11.68	7.72 $\pm$ 0.85
Model	242.72 $\pm$ 10.92**	210.24 $\pm$ 16.21**	9.43 $\pm$ 1.59**
Positive control	201.31 $\pm$ 8.58##	166.01 $\pm$ 6.13##	8.02 $\pm$ 0.88##
VEKAM	239.38 $\pm$ 12.61	203.70 $\pm$ 14.87	9.39 $\pm$ 1.53
VEKBM	217.35 $\pm$ 11.83#	164.23 $\pm$ 12.58##	7.36 $\pm$ 0.55##
VEKCM	198.79 $\pm$ 9.54##	154.63 $\pm$ 17.69##	7.14 $\pm$ 0.50##
VEKDM	237.86 $\pm$ 6.11#	221.25 $\pm$ 15.41	9.01 $\pm$ 2.02

VEKA, VEKB, VEKC and VEKD: fractions A, B, C and D of kansui fry-baked with vinegar.

VEKAM, VEKBM, VEKCM and VEKDM: Model rats supplemented with VEKA, VEKB, VEKC and VEKD.

* $p < 0.05$, ** $p < 0.01$ compared to control group.

$p < 0.05$, ## $p < 0.01$ compared to model group.

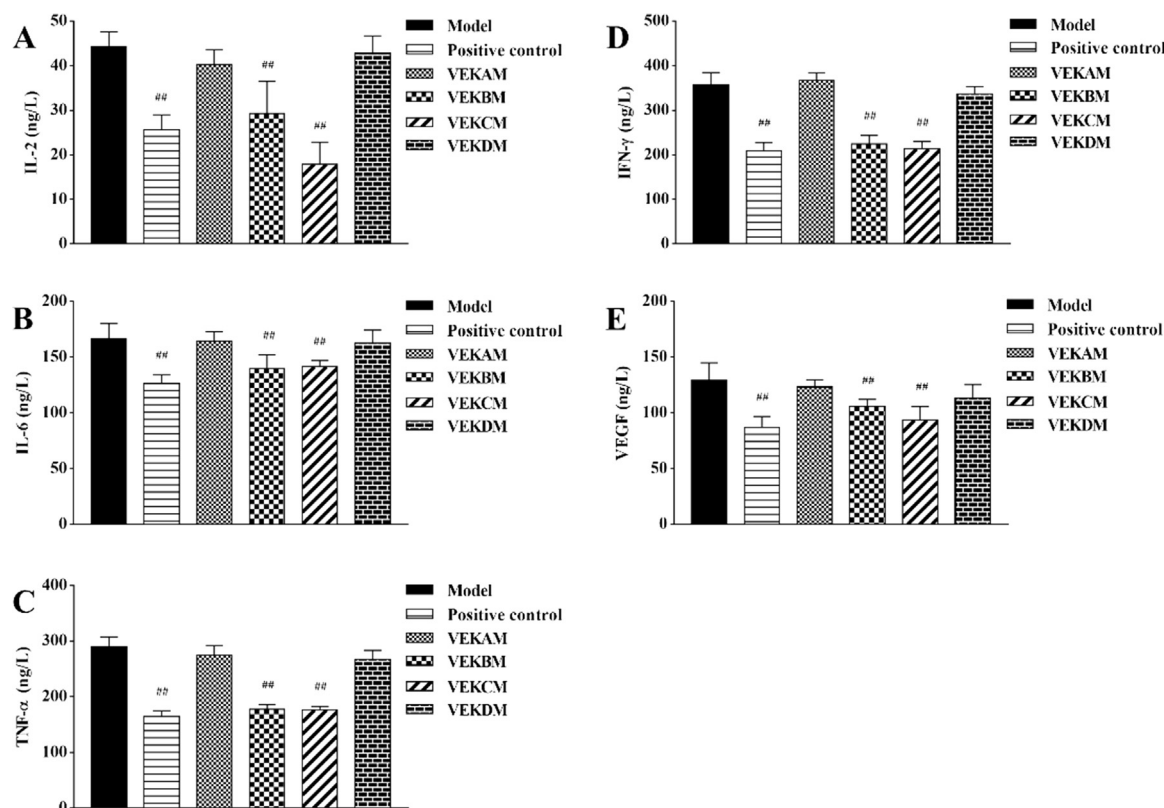


Fig. 2. Effects of VEKF on the pro-inflammatory cytokines in ascites using ELISA. (A) IL-2, (B) IL-6, (C) TNF- α , (D) IFN- γ , (E) VEGF. Data are presented as mean $\pm$ SD (n = 8). # $p < 0.05$, ## $p < 0.01$ compared to model group.

Fig. S2–S4. Compared with the control group, the liver NF- κ B ($P < 0.05$, Fig. 4A) and ICAM-1 ($P < 0.01$, Fig. 4B) levels of rats in the VEKBC group were increased remarkably, while the liver E-cadherin level was significantly lowered ($P < 0.01$, Fig. 4C). And the liver ICAM-1 level in the VEKCC group was increased obviously ($P < 0.01$, Fig. 4B), while the liver E-cadherin level was remarkably reduced ($P < 0.05$, Fig. 4C). Compared with the control group, as similar as VEKB and VEKC performed in the liver, these two fractions also could cause obvious change of NF- κ B, ICAM-1 and E-cadherin in the stomach and small intestine of normal rats. The results indicated that VEKB and VEKC could cause inflammatory damage in the liver, stomach and small intestine. Moreover, in the three organs, NF- κ B and ICAM-1 were significantly increased in the model group relative to the control group ($P < 0.01$), while the level of E-cadherin dramatically inhibited ($P < 0.01$). The data demonstrated that the successful establishment of MAE rats would with the inflammatory damage in the liver and gastrointestinal tract. After treating with VEKB and VEKC, these indexes in the three organs were all improved obviously except the stomach ICAM-

1 in the VEKBM group. These results showed that VEKB and VEKC could cause inflammatory damage in the liver and gastrointestinal tract, but also could reduce the similar injuries in model rats.

3.3.3. Effects of VEKF on apoptosis in rat tissues

3.3.3.1. Effects of VEKF on apoptosis index in rat tissues. The effects of VEKF on apoptosis index (AI) in the liver, stomach and small intestine from different groups were investigated by using the situ cell death detection kit (Fig. 5). The detailed information is shown in Table S6 and Fig. S5. Compared with the control group, the liver AI of normal rats were increased remarkably with the supplemented of VEKB ($P < 0.01$), VEKC ($P < 0.05$) and VEKD ($P < 0.05$). Similarly, the stomach and small intestine AI in the VEKBC and VEKCC groups were also significantly increased compared to that of control group. Additionally, in these three organs, the AI in the model group were obviously higher than that in the control group ($P < 0.01$). The hepatic and gastrointestinal injuries in model rats were further proved by these data. And the apoptotic cells were reduced after drugs treatment.

Table 3

Effects of VEKF on PRA, the serum Ang II, ALD and ADH levels (mean $\pm$ SD, n = 8).

Groups	PRA (ng/L)	Ang II (ng/L)	ALD (ng/L)	ADH (ng/L)
Control	112.71 $\pm$ 13.83	106.26 $\pm$ 14.10	271.32 $\pm$ 12.41	6.49 $\pm$ 1.06
Model	174.59 $\pm$ 24.50**	220.24 $\pm$ 17.97**	428.29 $\pm$ 17.72**	15.61 $\pm$ 2.54**
Positive control	134.82 $\pm$ 17.59##	164.65 $\pm$ 16.70##	363.12 $\pm$ 18.13##	8.47 $\pm$ 1.38##
VEKAM	165.31 $\pm$ 29.19	215.64 $\pm$ 14.24	402.73 $\pm$ 24.24	14.97 $\pm$ 1.35
VEKBM	157.43 $\pm$ 13.31#	183.98 $\pm$ 12.99##	340.67 $\pm$ 12.42##	10.27 $\pm$ 1.73#
VEKCM	149.37 $\pm$ 14.85##	179.52 $\pm$ 17.45##	317.12 $\pm$ 17.79##	8.81 $\pm$ 1.05##
VEKDM	162.69 $\pm$ 19.03	214.75 $\pm$ 12.46	422.93 $\pm$ 14.82	15.09 $\pm$ 2.67

VEKA, VEKB, VEKC and VEKD: fractions A, B, C and D of kansui fry-baked with vinegar.

VEKAM, VEKBM, VEKCM and VEKDM: Model rats supplemented with VEKA, VEKB, VEKC and VEKD.

* $p < 0.05$, ** $p < 0.01$ compared to control group.

$p < 0.05$, ## $p < 0.01$ compared to model group.

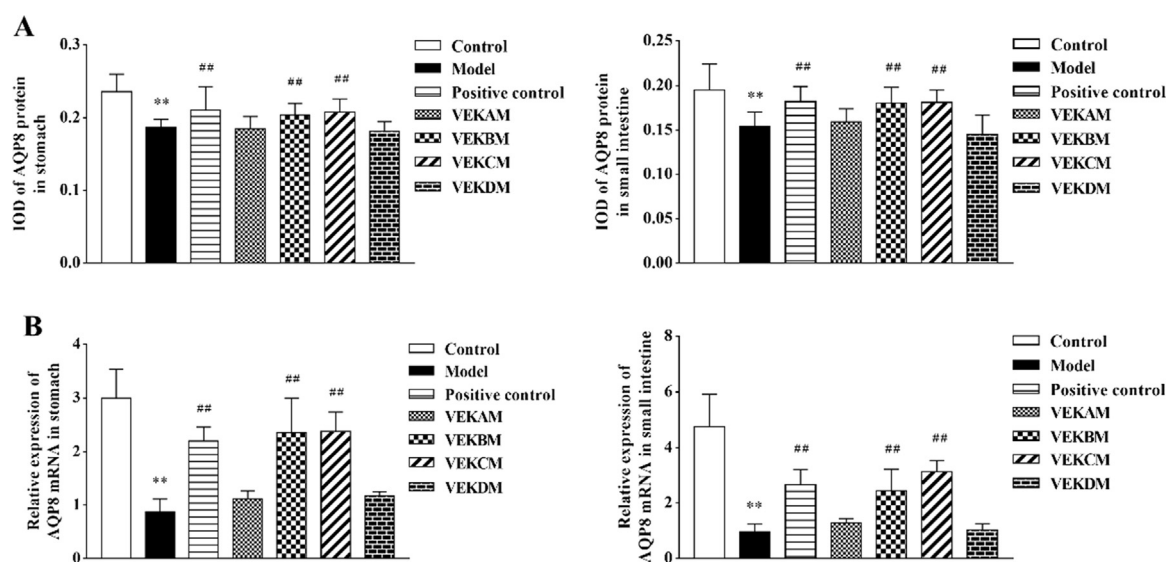


Fig. 3. Effects of VEKF on AQP8 protein expression in the stomach and small intestine using quantitative immunohistochemistry (A), and Real-time PCR of mRNA (B). Data are presented as mean $\pm$ SD (n = 8). * p < 0.05, ** p < 0.01 compared to control group. # p < 0.05, ## p < 0.01 compared to model group.

Table 4

Effects of VEKF on the serum AST and ALT levels (mean $\pm$ SD, n = 8).

Groups	AST (U/L)	ALT (U/L)
Control	9.81 $\pm$ 2.14	17.53 $\pm$ 5.16
Model	27.73 $\pm$ 8.75**	58.04 $\pm$ 11.20**
VEKAC	12.51 $\pm$ 2.49	20.99 $\pm$ 6.06
VEKBC	15.81 $\pm$ 3.90**	31.66 $\pm$ 6.40**
VEKCC	12.53 $\pm$ 2.36	29.93 $\pm$ 4.89**
VEKDC	12.12 $\pm$ 2.60	23.17 $\pm$ 3.26
VEKAM	27.96 $\pm$ 5.02	55.76 $\pm$ 7.64
VEKBM	19.20 $\pm$ 5.80##	45.11 $\pm$ 4.34##
VEKCM	18.56 $\pm$ 2.96##	49.72 $\pm$ 2.04##
VEKDM	34.34 $\pm$ 3.03	60.77 $\pm$ 6.18

VEKA, VEKB, VEKC and VEKD: fractions A, B, C and D of kansui fry-baked with vinegar.

VEKAC, VEKBC, VEKCC and VEKCD: Normal rats supplemented with VEKA, VEKB, VEKC and VEKD.

VEKAM, VEKBM, VEKCM and VEKDM: Model rats supplemented with VEKA, VEKB, VEKC and VEKD.

* p < 0.05, ** p < 0.01 compared to control group.

p < 0.05, ## p < 0.01 compared to model group.

Table 5

Effects of VEKF on the serum GSH, MDA, LDH and SOD levels (mean $\pm$ SD, n = 8).

Groups	GSH (μ mol/L)	MDA (nmol/mL)	LDH (U/mL)	SOD (U/mL)
Control	83.93 $\pm$ 7.76	6.74 $\pm$ 0.68	3.72 $\pm$ 1.21	64.80 $\pm$ 4.87
Model	33.32 $\pm$ 4.48**	16.22 $\pm$ 1.52**	18.16 $\pm$ 2.78**	34.14 $\pm$ 11.62**
VEKAC	79.81 $\pm$ 9.25	6.76 $\pm$ 1.05	3.69 $\pm$ 0.66	63.20 $\pm$ 5.82
VEKBC	69.53 $\pm$ 11.66*	11.44 $\pm$ 1.56**	5.12 $\pm$ 0.96**	49.03 $\pm$ 7.45**
VEKCC	72.96 $\pm$ 13.08	8.03 $\pm$ 0.96*	4.48 $\pm$ 1.03	51.05 $\pm$ 6.25**
VEKDC	80.69 $\pm$ 10.39	7.14 $\pm$ 0.68	4.30 $\pm$ 0.90	62.30 $\pm$ 5.70
VEKAM	31.83 $\pm$ 9.95	14.51 $\pm$ 1.11	15.33 $\pm$ 2.34	37.15 $\pm$ 6.80
VEKBM	42.16 $\pm$ 5.39#	11.79 $\pm$ 1.57##	10.50 $\pm$ 1.55##	51.87 $\pm$ 6.43##
VEKCM	51.07 $\pm$ 6.42##	10.30 $\pm$ 1.83##	10.48 $\pm$ 1.64##	53.87 $\pm$ 5.94##
VEKDM	34.60 $\pm$ 7.72	14.57 $\pm$ 1.27#	16.69 $\pm$ 2.24	42.13 $\pm$ 4.68

VEKA, VEKB, VEKC and VEKD: fractions A, B, C and D of kansui fry-baked with vinegar.

VEKAC, VEKBC, VEKCC and VEKCD: Normal rats supplemented with VEKA, VEKB, VEKC and VEKD.

VEKAM, VEKBM, VEKCM and VEKDM: Model rats supplemented with VEKA, VEKB, VEKC and VEKD.

* p < 0.05, ** p < 0.01 compared to control group.

p < 0.05, ## p < 0.01 compared to model group.

Compared with the model group, in the three organs, the AI in MAE rats were remarkably decreased with the treatment of VEKB and VEKC. The results further indicated that VEKB and VEKC have hepatic and gastrointestinal toxicity. And these two fractions showed protect function not toxicity in model rats.

3.3.3.2. Effects of VEKF on the apoptotic signaling in rat tissues. The expression of some critical apoptosis signals (caspase-3, caspase-8, caspase-9, Bax and Bcl-2) in the liver, stomach and small intestine of all ten groups was established by immunohistochemical staining (Fig. 6). The detailed information is shown in Table S7–S11 and Fig. S11–S10. Compared with the control group, the expression of liver caspases (caspase-3 and 8, P < 0.01, Fig. 6A–B; caspase-9, P < 0.05, Fig. 6C) and Bax (P < 0.05, Fig. 6D) of rats in the VEKBC group were significantly enhanced, while the liver Bcl-2 was reduced remarkably (P < 0.05, Fig. 6E). Additionally, the expression of liver caspase-3 (P < 0.01, Fig. 6A), caspase-8 (P < 0.05, Fig. 6B) and Bax (P < 0.05, Fig. 6D) in the VEKCC group were remarkably augmented. Compared with the control group, as similar as VEKB and VEKC performed in the liver, these two fractions also could cause most of the five factors obvious change in the stomach and small intestine of normal rats (all P < 0.05). Compared with the control group, the expression of stomach Bax in the VEKDC group also was remarkably increased

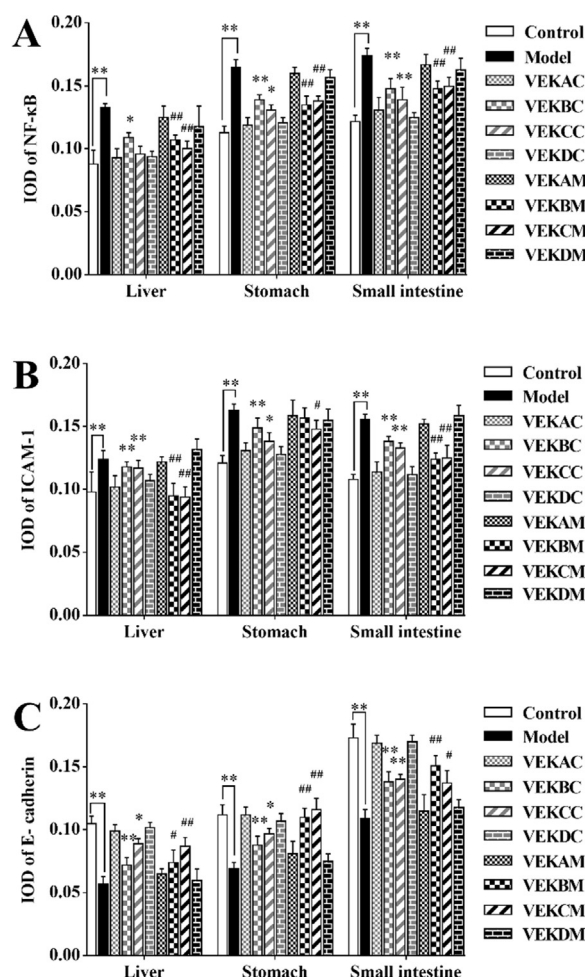


Fig. 4. Effects of VEKF on the inflammatory parameters in the liver, stomach and small intestine using quantitative immunohistochemistry. (A) NF-κB, (B) ICAM-1, (C) E-cadherin. Data are presented as Mean $\pm$ SD ($n = 8$). * $p < 0.05$, ** $p < 0.01$ compared to control group. # $p < 0.05$, ## $p < 0.01$ compared to model group.

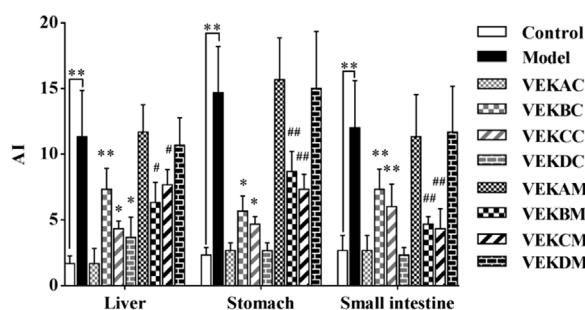


Fig. 5. Effects of VEKF on the AI in the liver, stomach and small intestine assessed by TUNEL. The AI is expressed as the percentage of apoptotic positive nuclei among at least 100 cells counted under a fluorescence microscope (magnification, 400 $\times$). Data are presented as mean $\pm$ SD ($n = 8$). * $p < 0.05$, ** $p < 0.01$ compared to control group. # $p < 0.05$, ## $p < 0.01$ compared to model group.

($P < 0.05$). The hepatic and gastrointestinal toxicity of VEKB and VEKC were further elucidated by the further investigations of apoptosis. Furthermore, in the three organs, the expression of caspase-3, caspase-8, caspase-9 and Bax were obviously activated in the model group relative to the control group (all $P < 0.01$), while the expression of Bcl-2 dramatically inhibited (all $P < 0.01$). After treating with VEKB and VEKC, the expression of these apoptosis signals in the three organs were

all improved significantly (all $P < 0.05$) except the liver caspase-9 in the VEKBM group. Compared with the model group, the expression of caspase-9 in the liver ($P < 0.05$) and small intestine ($P < 0.05$) were significantly decreased ($P < 0.05$) in the VEKD group. Based on the preliminary investigations of apoptotic mechanism, these results further revealed the hepatic and gastrointestinal toxicity of VEKB and VEKC. Additionally, these two fractions showed protect function not toxicity in model rats.

4. Discussion

As the well-known toxic traditional Chinese medicine, kansui exhibited a drastic hydragogue effect, and were widely applied for centuries in China. Stipulated in Chinese Pharmacopoeia, kansui must be fry-baked with vinegar for oral administration. In the present study, results of efficacy evaluation demonstrated that VEKB and VEKC treatment could promote urination by suppressing the renin-angiotensin-aldosterone system (RAAS), decrease the ascites. Meanwhile, the electrolyte and acid-basic disturbances of model rat were also regulated. Furthermore, the ascites pro-inflammatory cytokines in model rats were obviously decreased by treated with VEKB and VEKC, while the expression of gastrointestinal AQP8 protein was notably enhanced. In the toxicity evaluation, VEKB and VEKC showed obvious hepatic and gastrointestinal toxicity in normal rats. And relative to VEKC, VEKB was speculated with the potentiality of stronger toxicity on normal rats. Interestingly, these two fractions improved different toxicity assessment indexes instead of moving to the worse on model rats. This study demonstrated that VEKB and VEKC may be promising candidates for malignant ascites effusion treatment. And this may show more hepatic and gastrointestinal protection rather than the toxicity in the treatment of ascites.

Mostly, malignant ascites effusion is the pathologic accumulation of fluid in the peritoneal cavity caused by primary or metastatic tumors (e.g. ovarian, breast, liver and colon carcinomas), particularly in the advanced stages of cancer. Notably, the model rats showed oliguria with substantive ascites compared with the control group. In this experiment, VEKB and VEKC could obviously promote urination and evacuate the ascites. Commonly, the resorption of sodium and water will increase when the kidney produces urine, while the potassium excretion is enhanced. Therefore, the diuretics which act through the renin-angiotensin-aldosterone system (RAAS) and renal sodium handling system play a critical role in the treatment of ascites (Lucena et al., 2002; Moore and Aithal, 2006; Runyon, 2009). As the end of the RAAS, aldosterone (ALD) promotes kidney sodium and water resorption, subsequently leads to urine reduction (Kuiper et al., 2008). And ALD is regulated by rein and angiotensinogen. Angiotensinogen, the hepatic product, could be catalytically converted to angiotensin I (Ang I) by renin. And plasma renin activity (PRA) was usually detected to measure the ability to form Ang I in plasma from endogenous angiotensinogen and renin (Plouin et al., 1990). Next, some enzymes act on Ang I resulting in a sequential synthesis of angiotensin II (Ang II) and ALD. Furthermore, Ang II also could augment sodium reabsorption by the proximal tubules and stimulate the release of anti-diuretic hormone (ADH). Thus, the levels of renin, Ang II, ALD and ADH in the plasma are positively related. In this experiment, PRA, serum Ang II, ALD and ADH levels were remarkably inhibited in VEKBM and VEKCM groups compared with the model group. Meanwhile, the urinary electrolyte and acid-basic disturbances in model rats also could be significantly regulated by VEKB and VEKC. Additionally, it has been investigated that aquaporin water channels (AQPs) represent a major transcellular route for water transport in the gastrointestinal tract (Zhu et al., 2016). And AQPs play non-negligible functions in the formation and remission of ascites (Leiva et al., 2007). Compared with the model group, the expressions of the stomach and small intestine AQP8 were significantly enhanced in the VEKBM and VEKCM groups (Fig. 3). And the experiment of Real-time PCR of AQP8 mRNA also indirectly demonstrated

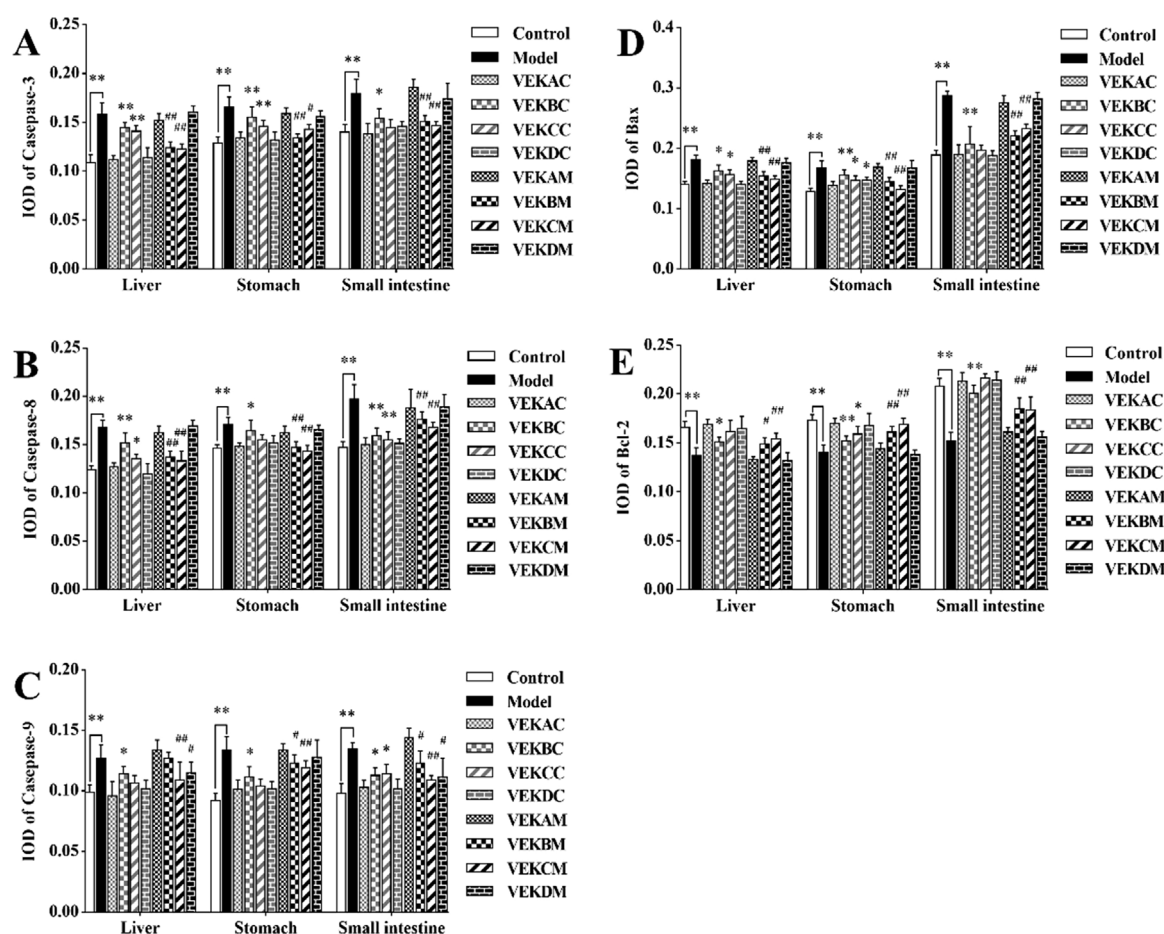


Fig. 6. Effects of VEKF on the apoptotic signaling in the liver, stomach and small intestine using quantitative immunohistochemistry. (A) caspase-3, (B) caspase-8, (C) caspase-9, (D) Bax, (E) Bcl-2. Data are presented as mean $\pm$ SD ($n = 8$). * $p < 0.05$, ** $p < 0.01$ compared to control group. # $p < 0.05$, ## $p < 0.01$ compared to model group.

these results. The above results indicated that these two fractions may promote urination by suppressing the RAAS. Additionally, [Arroyo and Colmenero \(2003\)](#) have reviewed that the activation of the RAAS is related to the development of ascites in cirrhosis. Thus, we speculate that the decrease of malignant ascites may partly due to the suppression on the RAAS. That is what we could further research. And the enhanced expression of AQP8 protein in the stomach and small intestine may also be associated with the efficacy of these two fractions.

Furthermore, the detection of ascites pro-inflammatory cytokines (TNF- α , IFN- γ , etc.) levels is a relative simple and effective manner to evaluate the seriousness of this disease, which has been reported in several studies ([Lu, 2011](#); [Nascimento et al., 2004](#); [Sharma et al., 2006](#); [Sipaksmigiel et al., 2017](#); [Such et al., 2001](#); [Zhao et al., 2016](#)). The milder the disease be, the lower these factors levels. As shown in [Fig. 2](#), the levels of the five mediators in the ascites of model rats also were notably decreased under the supplement of VEKB and VEKC. Thus, we assume that VEKB and VEKC may reduce the produce of ascites by inhibiting inflammation. Though it still should be further proved. So, it is conferred that VEKB and VEKC may play a major role in the therapeutic effect of VEK. And the safety of medicines is an essential part of patient safety. A further research of VEKF's toxic effects is needed.

In recent studies, the oxidative damage was regarded as an important mechanism of organ injuries ([Liu et al., 2016](#); [Shang et al., 2016](#)). Generally speaking, the body's own oxidants and antioxidants all play a nonnegligible role in this process. On the one hand, as oxidants, excessive free radicals usually cause cell membranous lipid peroxidation injury resulting in the increase of cytoplasmic markers (most generally LDH) in the blood stream. Simultaneously, MDA, the crucial

degradation product of lipid peroxides generated from free radical attack, could cause secondary damage to cellular membrane ([Tietz et al., 1986](#)). On the other hand, the body's own direct or indirect free radical scavengers, which are collectively called antioxidants (e. g. GSH, SOD, etc.) also commonly show low levels when there is an oxidative damage ([Maier and Chan, 2002](#); [Meister, 1988](#)). In this experiment, especially in VEKBC group, the serum MDA and LDH levels showed an obvious increase compared with that of normal group, while the serum GSH and SOD levels were significantly decreased, indicating VEKB could cause serious oxidative damage. Additionally, [Cao et al. \(2015b\)](#) also found that the oxidative damage might be an important mechanism of the hepatic and gastrointestinal toxicity of kansui fry-baked with vinegar. It was speculated that VEKB might have serious organ toxicity due to the oxidative damage. Interestingly, in the VEKCC group, only two of above serum indexes obviously changed compared with that of normal group, which might show the differences of organ toxicity between VEKBC and VEKCC groups.

Furthermore, ALT and AST are generally used as sensitive evaluation indexes of hepatic injury ([Qi et al., 2016](#); [Sarhadynejad et al., 2016](#)). Hepatic injury usually manifests degeneration or necrosis of hepatocytes and results to these cytosolic enzymes leak into the blood stream, which causes serum enzymes increase. Compared with the normal group, VEKBC group noticeably increased concentrations of serum AST and ALT. But in the VEKCC group, only serum ALT obviously increased compared with that of normal group. ALT is mainly existed in cytoplasm of hepatocytes. And AST, which is mainly distributed in mitochondria and cytoplasm of hepatocytes. Moreover, lipid peroxidation could promote mitochondrial injury ([Liu and peng, 2005](#)).

Combined with the results of oxidative damage assessment, VEKB might cause greater damage to the mitochondria of hepatocytes because of its possible stronger hepatotoxicity. The toxicity comparison of VEKB and VEKC still should be further studied.

Commonly, inflammation is closely related to oxidative damage, imbalanced inflammation accompanied by oxidative damage. Oxidative damage could destroy the close connection and adhesion of the mucosal cells, cause the low expression of E-cadherin and the high expression of ICAM-1, then lead to the increased attacking opportunity of harmful substances, initiating inflammation (Hurd et al., 2012; Kim et al., 2008; Thews et al., 2009). In this experiment, VEKB and VEKC supplement obviously suppressed E-cadherin expression in the liver and gastrointestinal tract of normal rats, while ICAM-1 expression was promoted significantly, indicating these two fractions also can promote inflammatory damage occurs in the liver and gastrointestinal tract. And NF- κ B, which has a fundamental role in inflammation and other natural and pathological processes, has been widely implicated in human diseases and in animal models of diseases (Zhang et al., 2017). NF- κ B could be activated by multiple substances, like reactive oxygen species (ROS), the crucial factors of oxidative damage (Grande et al., 2010; Shi et al., 2015). Furthermore, activated NF- κ B regulates the expression of multiple inflammatory substances (e. g., cytokines, chemokines, adhesion molecules, etc.), so as to control the inflammatory processes. ICAM-1 and E-cadherin are the crucial members of adhesion molecules. In this study, VEKB and VEKC supplement significantly increased the amount of NF- κ B in the gastrointestinal tract of normal rats. Although NF- κ B's activity hasn't been proved in the experiment, Chen et al. (2013) found that kansui could activate NF- κ B signaling pathway. It is a bold vision that NF- κ B may be activated by VEKB and VEKC to control the expressions of ICAM-1 and E-cadherin in the liver and gastrointestinal tract.

Moreover, apoptosis is associated with some pharmacological activities of TCMS, which induced through the extrinsic (receptor-mediated) or intrinsic (mitochondria-mediated) apoptosis pathways (Chen et al., 2016; Gao et al., 2015; Yan et al., 2014). In this regard, VEKB and VEKC also caused significant apoptosis in the three tissues of normal rats (Fig. 5). And the two apoptosis pathways both have a relationship with the activation of caspase-3, which cleaves target proteins leading to apoptosis. The extrinsic apoptosis is through a sequential activation of caspase-8, and caspase-3, which induced by the ligation of death receptors with their ligands (e.g. TNF- α , TRAIL, etc.). On the other hand, it has been accepted that the inhibition of anti-apoptotic protein Bcl-2 and activation of pro-apoptotic protein Bax is a classical way to promote the release of cytochrome c and the formation of apoptosome complex (cytochrome c, caspase-9 and Apaf-1), finally enhances the intrinsic apoptosis (Li-Weber, 2013). And activated caspase-9 could activate pro-caspase-3. Thus, the Bcl-2/Bax protein ratio is a crucial element of apoptotic cell death. As shown in Fig. 6, VEKB could obviously elevate the activities of caspase-3, -8, -9 and Bax in the liver and gastrointestinal tract of normal rats, while Bcl-2 expression was inhibited notably, which indicated that VEKB induced apoptosis possibly via extrinsic pathway and intrinsic pathway. Nevertheless, some of these factors' variation of VEKCC group was not remarkable compared to that of VEKBC group. These results indicated VEKB may cause more serious apoptotic cell death, and then manifest stronger organ toxicity. Additionally, as the crucial factors of oxidative damage, excessive ROS also is regarded as a direct inductive factor of apoptosis (Efferth et al., 2007). The generation of ROS may increase mitochondrial permeability directly or indirectly, then release cytochrome c, leading to caspase activation and apoptosis. And the results of oxidative damage studies demonstrated excessive generation of ROS occurred in this process. Thus, we assume that VEKB and VEKC killed the cells by generation of ROS. However, their mechanisms regarding to how ROS is induced are still a worthy question to further explore.

As we all known, the safety of drugs is an important issue, and the toxicological studies usually are conducted in normal and pathological

model animals nowadays. In this regard, we also follow this principle in this research. Obviously, there were serious hepatic and gastrointestinal injuries in Walker-256 tumor-bearing rats through the histopathological observation. And the significant changes of above test indicators also help us to further confirm this. Interestingly, the results demonstrated that the obvious hepatic and gastrointestinal toxic VEKB and VEKC could significantly improve different toxicity assessment indexes instead of moving to the worse in model rats. In other words, these evidences further proved the remedial effects of VEKB and VEKC to model rats.

This study provides evidence that VEKB and VEKC may promote urination through suppressing the RAAS. These two fractions also remarkably reduced the malignant ascites. Meanwhile, the body electrolyte and acid-basic disturbances also were corrected. Of course, the curative effect of these two fractions also may partly due to the up-regulated expression of AQP8 protein in the gastrointestinal tract. In this study, the inflammation was along with the serious hepatic and gastrointestinal injuries in model rats. And it turned out that VEKB and VEKC also had obvious anti-inflammation on model rats. Furthermore, it has been reported that inflammation is related to cancer (Mantovani et al., 2008). The cancer cells promote inflammation response occurs in the body, and inflammation also be partly responsible for the growth of cancer cells (Gregory and Houghton, 2011; Mantovani et al., 2011). Vicious circle again and again, finally may cause the malignant ascites. It might be an another nonnegligible reason for ascites decrease, and we also are curious about the effect of kansui on solid tumor, which will be the focus of follow-up studies.

Despite the regular belief that the herbal remedies are natural and safe, there have been several adverse effects associated with herbal remedies reported in the literatures, such as the liver injury (Lee et al., 2011; Teschke et al., 2014). Xiao-Chai-Hu-Tang (syo-saiko-to in Japanese), the famous Chinese herbal products, which has been reported to induce hepatotoxicity because of its possible inappropriate utilization, overdose, intrinsic toxicity and so on (Hsu et al., 2006; Itoh et al., 1995). Once again it has suggested the importance of the reasonable application of drugs to us. And it needs to be emphasized that drugs are only used clinically in patients. In other words, the applications of drugs' toxicity are more meaningful in pathological state if the toxicity plays the positive role in the treatment. Although the near-unanimous effective VEKB and VEKC still showed obvious hepatic and gastrointestinal toxicities in normal rats. However, based on the reasonable application in clinic, it's possible to exert the maximum effect of the toxic drugs if the toxicity was clarified under the normal and pathological states.

Additionally, only VEKA didn't have an effect on the indicators, either in the efficacy or toxicity evaluation. It suggested that we could pay more attention on the other fractions in the future research. And an accidental founding was that the low toxic VEKD could significantly improve some indicators in MAE rats. Therefore, there may be more findings if VEKD could be further studied. Another interesting finding in the research was that VEKC's influence on the part of toxicity assessment indexes in normal rats was inferior to VEKB. Our previous study (Lou et al., 2017) also found that VEKC showed lower toxicity to zebrafish embryos. By the analysis of UFLC-Q-TOF/MS, ingenane-type and japhane-type diterpenoids were the main compounds detected in VEKB and VEKC, respectively (Lou et al., 2017). The relevant data information was shown in the Table S11–S13 and Fig. S11–S13. Moreover, it has been reported that the lower cytotoxicity of japhane-type diterpenoids relative to ingenane-type in several studies (Wang et al., 2002, 2003). Possibly, japhane-type diterpenoids are the effective and low toxic constituents in kansui, which deserving in-depth development research, relatively.

In this research, it should be noted the main focus is the preliminary investigation of the concept of *You Gu Wu Yun*. Although there still existed some deficiency, such as the confirmation of NF- κ B's activity. Despite all this, this study does suggest the possibility of the application

of *You Gu Wu Yun*. And the material basis of VEK's toxicity and efficacy on MAE was further explored.

5. Conclusion

Taken together, the current study indicated that VEKB and VEKC's anti-effusion on malignant ascites might partly due to the urine promoting. And these two herbal fractions might significantly promote urination through suppressing the RAAS. Then the enhanced AQP8 protein expression in the gastrointestinal tract also might be linked to the efficacy. Interestingly, although VEKB and VEKC caused hepatic and gastrointestinal damage to the normal rats, also reduced the same damage of the model. It tentatively elucidated the traditional Chinese medicine application principle: *You Gu Wu Yun*. And all of these findings add the accumulating evidence that suggests the mechanisms and the main chemical basis regarding the toxicity and efficacy of kansui fry-baked with vinegar.

Acknowledgments

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Author contributions

Jian-Wei Lou, Liang-Liang Cao, Qiao Zhang, Dong-Jing Jiang, Wei-Feng Yao, Bei-Hua Bao, Yu-Dan Cao, Yu-Ping Tang, Li Zhang, Kun Wang, Guan-Cheng Dai.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.jep.2018.03.010>.

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